STK4021 – Mandatory assignment, Fall 2026

Bayesian inversion of seismic data

NAME – STK4021

14 October 2026

Self-report

To be written by the student: how the work proceeded, what was difficult, time-consuming or fun, what was relatively easy, collaboration and sources used (including AI tools).

Part 1

1a)

Since $\mathbf{m}$ and $\boldsymbol{\varepsilon}$ are independent and Gaussian, $(\mathbf{m}, \mathbf{d})$ is a linear transformation of a Gaussian vector and therefore jointly Gaussian, with

$$\begin{aligned} E(\mathbf{d}) &= \mathbf{A}\boldsymbol{\mu}_m, \\ \mathrm{Cov}(\mathbf{d}) &= \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I} \;=:\; \boldsymbol{\Sigma}_d, \\ \mathrm{Cov}(\mathbf{d}, \mathbf{m}) &= \mathrm{Cov}(\mathbf{A}\mathbf{m} + \boldsymbol{\varepsilon}, \mathbf{m}) = \mathbf{A}\boldsymbol{\Sigma}_m, \end{aligned}$$

so

$$\begin{pmatrix} \mathbf{m} \\ \mathbf{d} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_m \\ \mathbf{A}\boldsymbol{\mu}_m \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_m & \boldsymbol{\Sigma}_m\mathbf{A}^\top \\ \mathbf{A}\boldsymbol{\Sigma}_m & \boldsymbol{\Sigma}_d \end{pmatrix} \right).$$

The conditional distribution of a partitioned Gaussian gives the posterior $\mathbf{m} \mid \mathbf{d} \sim N(\boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d})$ with

$$\begin{aligned} \boldsymbol{\mu}_{m|d} &= \boldsymbol{\mu}_m + \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}(\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m), \\ \boldsymbol{\Sigma}_{m|d} &= \boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}\mathbf{A}\boldsymbol{\Sigma}_m. \end{aligned}$$

The pointwise 95% credibility interval for m_i is $(\boldsymbol{\mu}_{m|d})_i \pm 1.96\sqrt{(\boldsymbol{\Sigma}_{m|d})_{ii}}$.

1b)

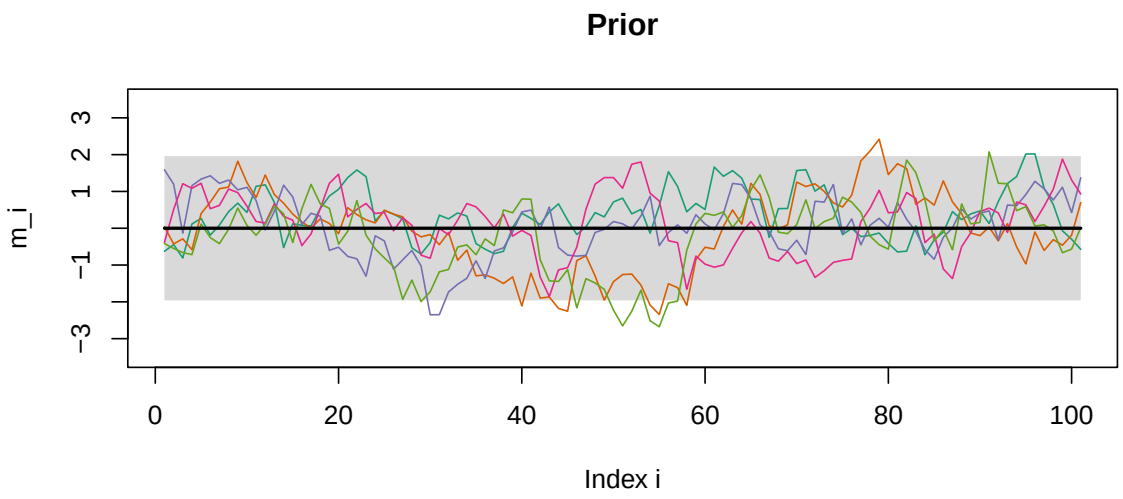


Figure 1: Prior expectation (black), 95% credibility interval (grey) and five simulated profiles.

Since $\Sigma_{m,ii} = 1$, the prior is $m_i \sim N(0, 1)$ and the interval is 0 ± 1.96 for all i. The profiles are simulated as $\boldsymbol{\mu}_m + \mathbf{L}\boldsymbol{x}$, with $\mathbf{L}\mathbf{L}^\top = \boldsymbol{\Sigma}_m$ (Cholesky) and $\boldsymbol{x} \sim N(\mathbf{0}, \mathbf{I})$.

1c)

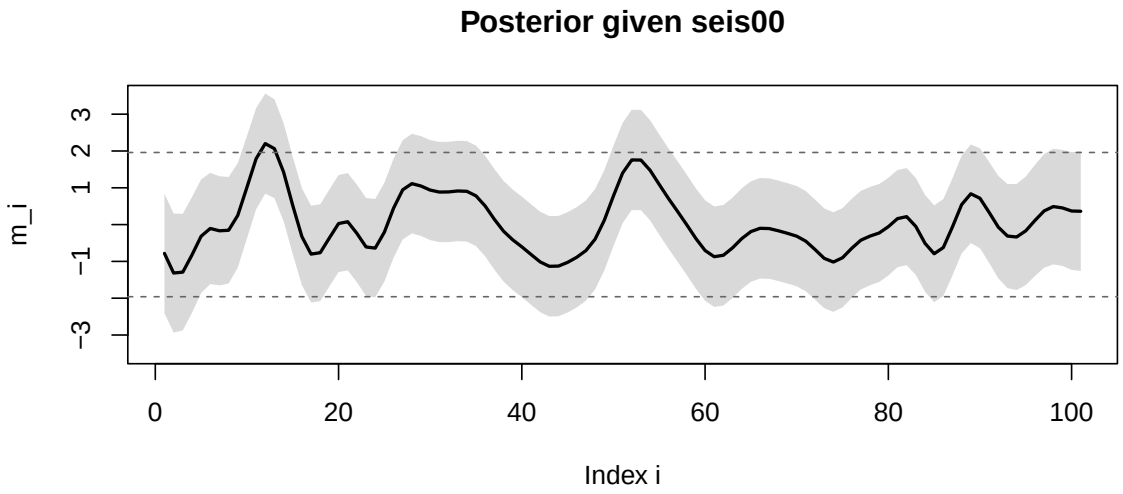


Figure 2: Posterior expectation (black) and 95% credibility interval (grey) given seis00. Dashed: prior interval.

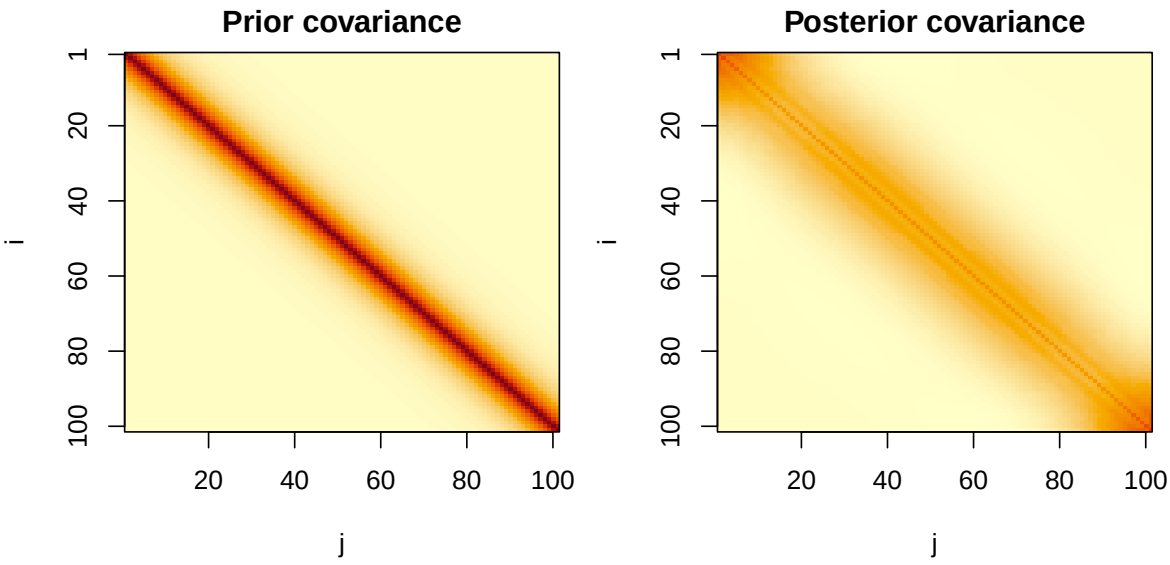


Figure 3: Prior (left) and posterior (right) covariance matrices, same colour scale.

The posterior standard deviations are 0.67–0.83, compared with 1 in the prior. The uncertainty is reduced, but much remains, and the posterior covariance is spread further from the diagonal. One reason is that $\mathbf{A1} = \mathbf{0}$ (the rows of $\mathbf{A}$ sum to zero). The data therefore contain no information about the overall level of $\mathbf{m}$, only about local changes.

1d)

Table 1: Posterior summaries of V ($B = 100$).

mean	median	sd
8.76	5.48	8.68

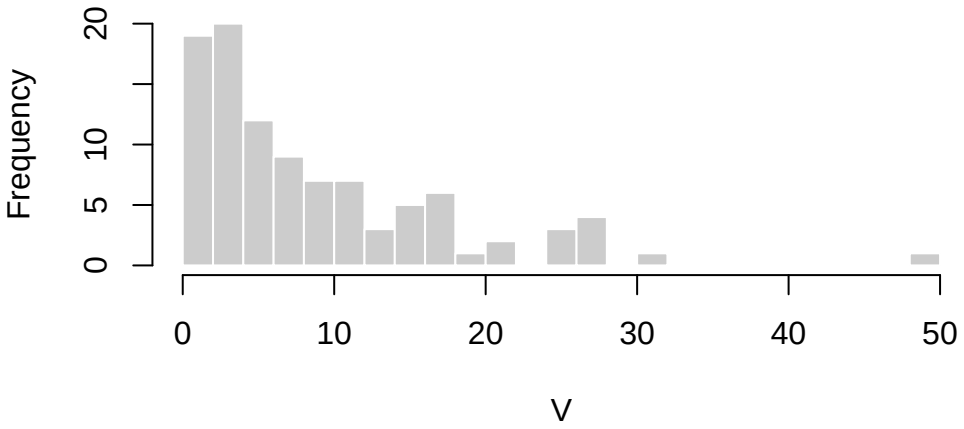


Figure 4: Histogram of 100 posterior samples of V.

We draw $\mathbf{m}^{(b)} \sim N(\boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d})$, $b = 1, \ldots, 100$, and compute $V^{(b)} = h(\mathbf{m}^{(b)})$. The posterior of V is right-skewed, so the mean is larger than the median.

1e)

Estimates from the ordered samples: $\widehat{P}_{10} = V_{(10)} = 0.92$ and $\widehat{P}_{90} = V_{(90)} = 20.64$.

The number of samples below P_{10} is $N \sim \text{Bin}(100, 0.1)$. Since $V_{(r)} \leq P_{10} < V_{(s)}$ exactly when $r \leq N \leq s - 1$,

$$P\big(V_{(r)} \leq P_{10} < V_{(s)}\big) = \sum_{k=r}^{s-1} \binom{100}{k} 0.1^k 0.9^{100-k}.$$

With $r = 5$ and $s = 17$ this probability is 0.956, which gives the interval $[V_{(5)}, V_{(17)}] = [0.19,\ 1.82]$ for P_{10}. The interval is wide compared with the estimate, so $B = 100$ is not sufficient for a precise P_{10}, and B should be increased.

1f)

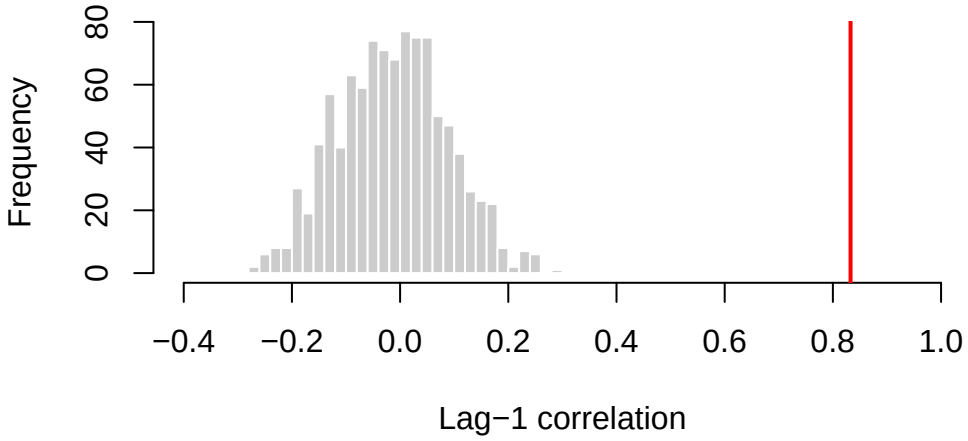


Figure 5: Lag-1 correlation for 1000 permutations of seis00 (histogram) and for the observed data (red line).

If the data were exchangeable, the observed order would be as likely as any permutation, and $\hat{\rho}_1$ would look like the permuted values. The observed lag-1 correlation is $\hat{\rho}_1 = 0.83$, while the permuted values lie in $[-0.28,\ 0.29]$. None of the 1000 permutations are as extreme ($p < 0.001$), so the claim of exchangeability is rejected. Neighbouring samples are strongly correlated, as expected from $\mathbf{d} = \mathbf{Am} + \boldsymbol{\varepsilon}$.

Appendix: R code

```
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE,
                      fig.align = "center", fig.width = 7.5, fig.height = 3.6,
                      out.width = "95%", fig.pos = "H")
A  <- as.matrix(read.table("data/A.txt"))   # forward operator (n x n)
dimnames(A) <- NULL
d0 <- scan("data/seis00.txt", quiet = TRUE)
d1 <- scan("data/seis01.txt", quiet = TRUE)
d2 <- scan("data/seis02.txt", quiet = TRUE)
n  <- nrow(A)
z  <- 1:n
# Prior covariance: exponential correlation function
exp_cov <- function(n, sigma2, ell) {
  D <- abs(outer(1:n, 1:n, "-"))           # matrix of |i - j|
  sigma2 * exp(-3 * D / ell)
}

# Posterior of m | d for d = A m + eps (Gaussian conditioning)
gauss_posterior <- function(d, A, mu, Sigma, sigma2_eps) {
  Sigma_d <- A %*% Sigma %*% t(A) + sigma2_eps * diag(nrow(A))   # Cov(d)
  K       <- Sigma %*% t(A) %*% solve(Sigma_d)                   # Cov(m,d) Cov(d)^-1
  mu_post <- drop(mu + K %*% (d - A %*% mu))
  S_post  <- Sigma - K %*% A %*% Sigma
  S_post  <- (S_post + t(S_post)) / 2      # make exactly symmetric (rounding)
  list(mean = mu_post, cov = S_post)
}

# Simulate B draws from N(mu, S): m = mu + L x, L L^T = S, x ~ N(0, I)
rmvn <- function(B, mu, S) {
  L <- t(chol(S + 1e-10 * diag(length(mu))))   # lower Cholesky factor (tiny jitter)
  X <- matrix(rnorm(length(mu) * B), length(mu), B)
  t(mu + L %*% X)                               # one draw per row
}

# Mean with pointwise 95% credibility band (and optional realisations)
band_plot <- function(mean, sd, main, ylim, sims = NULL) {
  lo <- mean - 1.96 * sd
  hi <- mean + 1.96 * sd
  plot(z, mean, type = "n", ylim = ylim, xlab = "Index i", ylab = "m_i", main = main)
  polygon(c(z, rev(z)), c(lo, rev(hi)), col = "grey85", border = NA)
  if (!is.null(sims)) matlines(z, t(sims), lty = 1, lwd = 1,
                               col = c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e"))
  lines(z, mean, lwd = 2)
}
mu <- 0; sigma2_m <- 1; ell <- 20; sigma2_eps <- 0.04
mu_m    <- rep(mu, n)
Sigma_m <- exp_cov(n, sigma2_m, ell)

set.seed(1)
prior_sims <- rmvn(5, mu_m, Sigma_m)
band_plot(mu_m, sqrt(diag(Sigma_m)), "Prior", ylim = c(-3.5, 3.5), sims = prior_sims)
post0 <- gauss_posterior(d0, A, mu_m, Sigma_m, sigma2_eps)
```

```r
sd0   <- sqrt(diag(post0$cov))
band_plot(post0$mean, sd0, "Posterior given seis00", ylim = c(-3.5, 3.5))
abline(h = c(-1.96, 1.96), lty = 2, col = "grey40")
cov_image <- function(S, main, zlim) {
  image(z, z, t(S)[, n:1], zlim = zlim, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "j", ylab = "i", main = main, axes = FALSE, useRaster = TRUE)
  axis(1); axis(2, at = n + 1 - c(1, 20, 40, 60, 80, 100), labels = c(1, 20, 40, 60, 80, 100))
  box()
}
zl <- range(Sigma_m, post0$cov)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
cov_image(Sigma_m, "Prior covariance", zl)
cov_image(post0$cov, "Posterior covariance", zl)
sd_range <- range(sd0)
A_one    <- max(abs(A %*% rep(1, n)))      # A applied to a constant vector
c_lev <- -1
h_V   <- function(m, c = c_lev) sum((c - m) * (m < c))   # exceedance volume

B <- 100
set.seed(2)
post_sims <- rmvn(B, post0$mean, post0$cov)
V <- apply(post_sims, 1, h_V)                              # one V per posterior draw

V_summary <- c(mean = mean(V), median = median(V), sd = sd(V))
knitr::kable(t(round(V_summary, 2)), caption = "Posterior summaries of $V$ ($B=100$).")
hist(V, breaks = 20, col = "grey80", border = "white", main = "", xlab = "V")
Vs  <- sort(V)                                          # order statistics
P10 <- Vs[10]
P90 <- Vs[90]

# N = number of samples below the true P10 ~ Bin(B, 0.1)
# V_(r) <= P10 < V_(s)  <=>  r <= N <= s - 1
r <- qbinom(0.025, B, 0.1)
s <- qbinom(0.975, B, 0.1) + 1
coverage <- pbinom(s - 1, B, 0.1) - pbinom(r - 1, B, 0.1)
lag1 <- function(x) cor(x[-1], x[-length(x)])   # lag-1 correlation

rho_obs <- lag1(d0)
set.seed(4)
rho_perm <- replicate(1000, lag1(sample(d0)))    # sample() gives a random permutation
n_extreme <- sum(abs(rho_perm) >= abs(rho_obs))

hist(rho_perm, breaks = 30, col = "grey80", border = "white", main = "",
     xlab = "Lag-1 correlation", xlim = c(-0.4, 1))
abline(v = rho_obs, col = "red", lwd = 2)
```